



Assignment 02

Data Mining

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Introduction

This report is the documentation of the assignment on Fp Growth, Class Association Rules (CAR) and Generalized Sequential Pattern (GSP) algorithm.

The dataset is called Online Retail II. It is a publicly available dataset under the common license act. It contains transactions for a UK-based, online retail store between December 2009 and December 2011. The online retail company sells unique gifts for all occasions. The dataset contains 1,067,371 rows x 8 columns. The attributes are

1. InvoiceNo: uniquely assigned 6 digit number for each transaction. If starting with C it means cancellation
2. Stock Code: uniquely assigned Product code
3. Description: the product name
4. Quantity: the amount of each product per transaction
5. InvoiceDate: Date and Time a transaction was made
6. UnitPrice: product price in sterling
7. CustomerID: unique number assigned to every customer
8. Country: the country where the customer is from

Methodology

Preprocessing

The first step was to load the csv file and display the head. This showed all 8 columns and their values.

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom



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Data Cleaning and Feature Engineering

To clean the data the cancelled transactions were removed. These were identified by their invoices which started with the letter C. Then null customer IDs and rows with either Quantity or Price that were null were removed.

As 92% of data was from customers from the United Kingdom, the rest was removed to reduce noise. All attributes were also type corrected if required.

Then a new column was created for Revenue by multiplying Quantity and Price.

Summary Statistics

```
Total invoices    : 33,541
Unique customers  : 5,350
Unique products   : 4,616
Date range        : 2009-12-01 to 2011-12-09
Avg basket size   : 20.57 items/invoice
```

Transaction Definition

To extract transactions from the dataset, the invoice number and stock code were grouped together. The unique set of these were the transactions, which were printed out as 33541.

Part A: FP Growth

First the FP Node and FP class were defined manually, without using libraries. The Node includes item, count, parent, children and link. The FP Tree included a constructor, link function, insert, ascend, mine, fp growth and association rule generator functions.

A0: Hand trace on 6 invoices

Six sample invoices were used to manually illustrate the global ordering and conditional pattern bases for fp growths understanding that is intuitive.

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```
A0: FP-Tree Walkthrough – 6 sample invoices
Invoice 489434: ['21232', '21523', '21871', '22041', '22064', '79323P', '79323W', '85048']
Invoice 489435: ['22195', '22349', '22350', '22353']
Invoice 489436: ['21181', '21333', '21754', '21755', '21756', '22107', '22109', '22111', '22119', '22142', '22194', '22295', '22296', '35004B']
Invoice 489437: ['10002', '20695', '20703', '20971', '21351', '21352', '21360', '21364', '21912', '21987', '21989', '22111', '22112', '22130']
Invoice 489438: ['20711', '21033', '21100', '21252', '21329', '21410', '21411', '84031A', '84031B', '84032A', '84032B', '84519A', '84519B', '']
Invoice 489440: ['22349', '22350']

Global item order desc: [('22349', 2), ('22350', 2), ('22111', 2), ('85048', 1), ('21232', 1), ('21523', 1), ('22064', 1), ('22041', 1)]
Conditional Pattern Base for item '22349':
Path: ['22350', '22353', '22195'] (count: 1)
Path: ['22350'] (count: 1)
Frequent patterns mined from these prefix paths reveal items that co-occur with '22349'.

Conditional Pattern Base for item '22350':
Path: ['22349', '22353', '22195'] (count: 1)
Path: ['22349'] (count: 1)
Frequent patterns mined from these prefix paths reveal items that co-occur with '22350'.
```

A1: Full FP Growth Run

The FP Growth algorithm was run on all transactions with minimum support at 2% and minimum confidence at 50%. This had the results that the total frequent itemsets were 221 and out of these 28 were 2 itemsets, while the rest were 1 itemsets.

A2: Rule Mining for Top 14 rules

When calculating rules the minimum confidence was set at 50% and was sorted by lift. The top 14 rules are as follows.

Top 15 Rules by Lift:				
Antecedent	Consequent	Sup	Conf	Lift
21231	21232	0.024	0.722	14.10
82494L	82482	0.029	0.565	11.77
82482	82494L	0.029	0.603	11.77
21755	21754	0.023	0.528	10.02
21977	21212	0.022	0.529	8.39
84991	21212	0.024	0.514	8.15
20726	20725	0.022	0.553	8.11
22384	20725	0.028	0.548	8.05
20727	20725	0.029	0.506	7.43
85099F	85099B	0.027	0.633	7.11
22386	85099B	0.029	0.602	6.76
85099C	85099B	0.023	0.552	6.20
21931	85099B	0.026	0.537	6.04
21733	85123A	0.033	0.707	5.04

A3: Proposed Bundles

Based on the rules above the proposed bundles are as follows:

Bundle 1: ['21231', '21232']

21231: SWEETHEART CERAMIC TRINKET BOX

21232: STRAWBERRY CERAMIC TRINKET BOX

Rule: {21231} → {21232}

Support=0.024 | Confidence=0.722 | Lift=14.10

Bundle 2: ['82494L', '82482']

82494L: WOODEN FRAME ANTIQUE WHITE

82482: WOODEN PICTURE FRAME WHITE FINISH

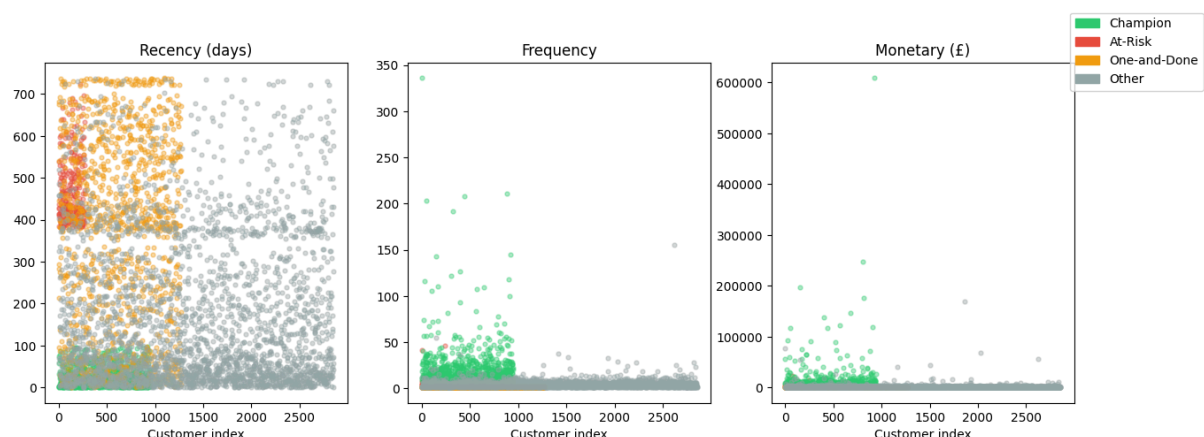
Rule: {82494L} → {82482}

Support=0.029 | Confidence=0.565 | Lift=11.77

Part B: Class Association Rules

Plots

The plots of recency, frequency and monetary values were created and evaluated side by side. At risk, champion and one and done customer segments were color coded, the rest were gray.





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Segment Summary

Segment summary (mean):			
	Recency	Frequency	Monetary
Segment			
At-Risk	470.7	4.9	1665.5
Champion	24.1	19.9	10237.3
One-and-Done	354.9	1.2	187.3
Other	167.5	4.1	1507.9

From the summary, we can see that At-Risk customers are the most recent while champions have the most monetary benefit and are the most frequent customers.

B1: Class Association Rules

Each segment was treated as a class label and CAR was done using minimum support at 1% and minimum confidence at 55%. The top 5 CARs per segment are shown below:



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Class Association Rules (minsup=1%, minconf=55%)

Champion (18948 invoices)

Antecedent	Class	Sup	Conf
{22746}	Champion	0.011	0.892
{22698, 22699}	Champion	0.016	0.884
{22423, 22698}	Champion	0.011	0.872
{22423, 22698}	Champion	0.011	0.868
{22697, 22698}	Champion	0.016	0.866

At-Risk (1329 invoices)

Antecedent	Class	Sup	Conf
{21900, 21901}	At-Risk	0.011	1.000
{21157, 21535}	At-Risk	0.010	1.000
{21527, 21531, 48185}	At-Risk	0.011	1.000
{21231, 22055}	At-Risk	0.012	1.000
{21755, 21756}	At-Risk	0.011	1.000

One-and-Done (1506 invoices)

Antecedent	Class	Sup	Conf
{22698}	One-and-Done	0.010	0.938
{22579}	One-and-Done	0.011	0.889
{22804}	One-and-Done	0.010	0.833
{21124}	One-and-Done	0.011	0.800
{22617}	One-and-Done	0.019	0.757

B2: New Customer Predictions

New customer prediction was made to see which segment they would fall in based on which 3 items they first ordered based on stock codes, based on Mined Class Association Rules.

Segment	Antecedent	Conf
Champion	{'20725', '20727', '22383'}	0.680

The segment predicted was champion.



Part c: Sequential Pattern Mining (GSP)

C1: Applying Generalized Sequential Patterns

A category dictionary was defined first then the category mapping was applied to the description which covered around 75% of values. Based on this 5350 customer sequences were made. Then with minimum support at 50% GSP was run.

Top 10 Frequent Length-2 Sequences are:

Sequence	Count	%
[General Homeware] → [General Homeware]	3528	65.9%
[Storage & Organisation] → [General Homeware]	3226	60.3%
[Stationery & Gifting] → [General Homeware]	3204	59.9%
[General Homeware] → [Storage & Organisation]	3187	59.6%
[General Homeware] → [Stationery & Gifting]	3183	59.5%
[Kitchen & Dining] → [General Homeware]	3177	59.4%
[General Homeware] → [Kitchen & Dining]	3108	58.1%
[General Homeware] → [Novelty & Decorative]	3046	56.9%
[Storage & Organisation] → [Storage & Organisation]	3028	56.6%
[Novelty & Decorative] → [General Homeware]	3022	56.5%

C2: Churn signal detection for At Risk Subpopulation Patterns

The results for the top 10 at risk customer sequences were found out.

Top 10 At-Risk Length-2 Sequences:

Sequence	Count	%
[General Homeware] → [General Homeware]	249	91.5%
[General Homeware] → [Storage & Organisation]	228	83.8%
[Storage & Organisation] → [General Homeware]	223	82.0%
[Kitchen & Dining] → [General Homeware]	222	81.6%



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[General Homeware] → [Kitchen & Dining]	218 80.1%
[Stationery & Gifting] → [General Homeware]	214 78.7%
[General Homeware] → [Stationery & Gifting]	210 77.2%
[Storage & Organisation] → [Storage & Organisation]	210 77.2%
[Storage & Organisation] → [Kitchen & Dining]	204 75.0%
[Kitchen & Dining] → [Kitchen & Dining]	201 73.9%

C3: Early Warning Trigger Rules

The 3 Rules found were as follows.

Trigger Rule Descriptions:

Rule 1:

TRIGGER : Customer's recent purchase event is in [Kitchen & Dining]

SEQUENCE : Kitchen & Dining → General Homeware

ACTION : Send discount voucher to pre-empt churn

Support : 222 at-risk customers match this full pattern

Confidence: 5.5% (245 of 4420 prefix-matching customers are At-Risk)

Estimated flags: 180 customers flagged per month

Rule 2:

TRIGGER : Customer's recent purchase event is in [Kitchen & Dining]

SEQUENCE : Kitchen & Dining → Kitchen & Dining

ACTION : Send discount voucher to pre-empt churn

Support : 201 at-risk customers match this full pattern

Confidence: 5.5% (245 of 4420 prefix-matching customers are At-Risk)



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Estimated flags: 180 customers flagged per month

Rule 3:

TRIGGER : Customer's recent purchase event is in [Kitchen & Dining]

SEQUENCE : Kitchen & Dining → Storage & Organisation

ACTION : Send discount voucher to pre-empt churn

Support : 200 at-risk customers match this full pattern

Confidence: 5.5% (245 of 4420 prefix-matching customers are At-Risk)

Estimated flags: 180 customers flagged per month

References

1. Dataset: archive.ics.uci.edu/dataset/502/online+retail+ii
2. FP-Growth paper: Han, J., Pei, J., Yin, Y. (2000). Mining frequent patterns without candidate generation. ACM SIGMOD.
3. GSP paper: Srikant, R., Agrawal, R. (1996). Mining sequential patterns: Generalizations and performance improvements. EDBT.